

Homework 1 – STAT 6611

Due Friday, September 18 by midnight (submit electronically/upload in gradescope under ‘Assignments’)

1. Let X , W , and Z denote matrices.

(a) If $\mathcal{C}(X) = \mathcal{C}(W)$, show that $P_X = P_W$, i.e., the projection matrices must be the same.

Hint: Consider $(P_W - P_X)'(P_W - P_X)$.

(b) If X is an $n \times n$ matrix of rank n , show $P_X = I_n$ (i.e., the $n \times n$ identity matrix).

(c) Let $\mathcal{C}^\perp(X)$ denote the orthogonal complement of $\mathcal{C}(X)$ (i.e., $\underline{y} \in \mathcal{C}^\perp(X)$ if and only if $\underline{x}'\underline{y} = 0$ for any column $\underline{x}$ of X). If a vector $\underline{b} \in \mathcal{C}(X)$ and $\underline{b} \in \mathcal{C}^\perp(X)$, show that $\underline{b}$ must equal the zero vector (this also implies that a non-zero vector in $\mathcal{C}(X)$ must be linearly independent from a non-zero vector in $\mathcal{C}^\perp(X)$).

Hint: Write $\underline{b} = X\underline{a}$ and consider $\underline{b}'\underline{b}$.

(d) Let X be an $n \times p$ matrix of rank $r \geq 1$, and Z be an $n \times (n - r)$ matrix with columns formed by $n - r$ linearly independent vectors in $\mathcal{C}^\perp(X)$. Show that $P_Z = I_n - P_X$.

Hint: Put r linearly independent columns of X along with the columns of Z into one $n \times n$ matrix $\tilde{X}$ (if $n = r$, technically Z is just a zero vector). Compute $P_{\tilde{X}}$ and use (a)-(b).

(e) For a vector $\underline{y}$, show that $P_X \underline{y} = 0$ if and only if $\underline{y} \in \mathcal{C}^\perp(X)$.

Hint: use (d) and multiply $I = I - P_X + P_X$ by $\underline{y}$

(f) Show that $P_X + P_W$ is a projection matrix if and only if $X'W = 0$ (the columns of X and W must be orthogonal).

Hint: If $P_X + P_W$ is a projection matrix, one can show $P_X P_W = -P_W P_X$ must hold; this relation implies that any vector $\underline{y}$ that is in both $\mathcal{C}(X)$ and $\mathcal{C}(W)$ must be zero (why?). Further, for $M = P_X P_W$, the relation also gives $M' M = P_X P_W P_W$ and $M' M = P_W P_X P_W$, meaning any column of $M' M$ belongs to $\mathcal{C}^\perp(X)$ and to $\mathcal{C}^\perp(W)$.

(g) Show that $P_X - P_W$ is a projection matrix if and only if $\mathcal{C}(W) \subset \mathcal{C}(X)$.

Hint: If $P_X - P_W$ is a projection matrix then $(I - P_X)P_W = -P_W(I - P_X)$ must hold; then the proof is kind of the same as in (f), using that (from (d)) $P_Z = I_n - P_X$ holds, where Z is a matrix with linearly independent columns that span $\mathcal{C}^\perp(X)$.

2. Suppose that $\mathcal{C}(C)$ is a subspace of $\mathcal{C}(A)$ and that $\mathcal{R}(D)$ is a subspace of $\mathcal{R}(A)$.

(a) Show that $AA^-C = C$ and $DA^-A = D$.

(b) Using (a), show that

$$\begin{bmatrix} I & 0 \\ -DA^- & I \end{bmatrix} \begin{bmatrix} A & C \\ D & B \end{bmatrix} \begin{bmatrix} I & -A^-C \\ 0 & I \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & Q \end{bmatrix},$$

where $Q = B - DA^-C$, and I denotes the identity matrix of appropriate dimensions.

(c) Using (b) with the fact that the two matrices defined by “ I on the diagonals” are square and invertible in (b), show that

$$\begin{bmatrix} I & -A^-C \\ 0 & I \end{bmatrix} \begin{bmatrix} A^- & 0 \\ 0 & Q^- \end{bmatrix} \begin{bmatrix} I & 0 \\ -DA^- & I \end{bmatrix}$$

is a generalized inverse of

$$M \equiv \begin{bmatrix} A & C \\ D & B \end{bmatrix}$$

- (d) Show that the above generalized inverse matches that given in (i) of Theorem 1.4.
- (e) Show that the conditions imposed on the matrices B , C , and D in part (ii) of Theorem 1.4 are satisfied if there exists a matrix $Z = (Z_1, Z_2)$ for which

$$M = Z'Z = (Z_1, Z_2)'(Z_1, Z_2) = \begin{bmatrix} Z_1'Z_1 & Z_1'Z_2 \\ Z_2'Z_1 & Z_2'Z_2 \end{bmatrix} = \begin{bmatrix} A & C \\ D & B \end{bmatrix}.$$

3. For each of the following statements, indicate whether the statement is true or false, and explain.

- (a) If $Ax = y$ has a unique solution, then A must be a nonsingular matrix.
- (b) If G is a generalized inverse of A , then $GA = I$, where I denotes the identity matrix of appropriate dimensions.
- (c) If A has a generalized inverse G such that $GA = I$, then A must be a (square) nonsingular matrix.
- (d) If A is a symmetric matrix, then all its generalized inverses are also symmetric.
- (e) If A is a symmetric matrix, then it has at least one symmetric generalized inverse.

4. For each of the following questions, give, if possible, a 3×3 **nondiagonal** matrix A that has the stated property.

- (a) A is a generalized inverse of A .
- (b) A' is a generalized inverse of A .
- (c) The identity matrix I is a generalized inverse of A .
- (d) Every 3×3 matrix is a generalized inverse of A .
- (e) Every generalized inverse of A is nonsingular.

5. For any matrix A , let B and C be any two matrices that satisfy $A'AB = A'$ and $AA'C' = A$ (the existence of such B and C follows from Corollary 2 to Lemma 1.4). Let $G = CAB$. Show that G satisfies the following four conditions:

- (i) $AGA = A$, (ii) $GAG = G$, (iii) $(GA)' = GA$, and (iv) $(AG)' = AG$.